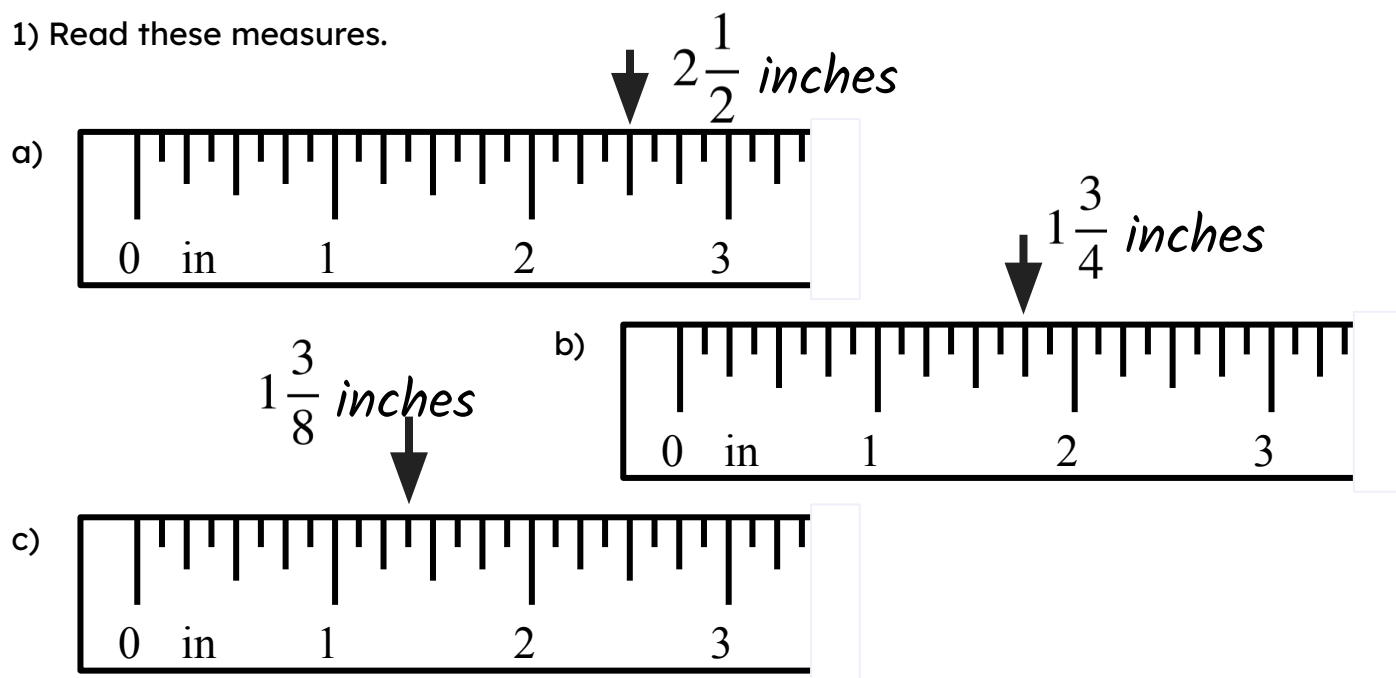




# Engineer

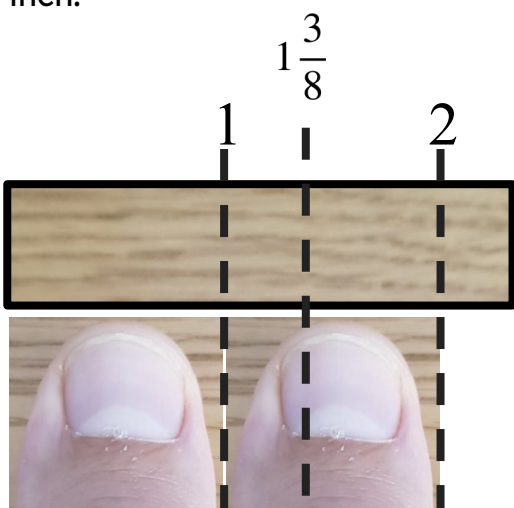
## Task A: Precision in engineering

1) Read these measures.

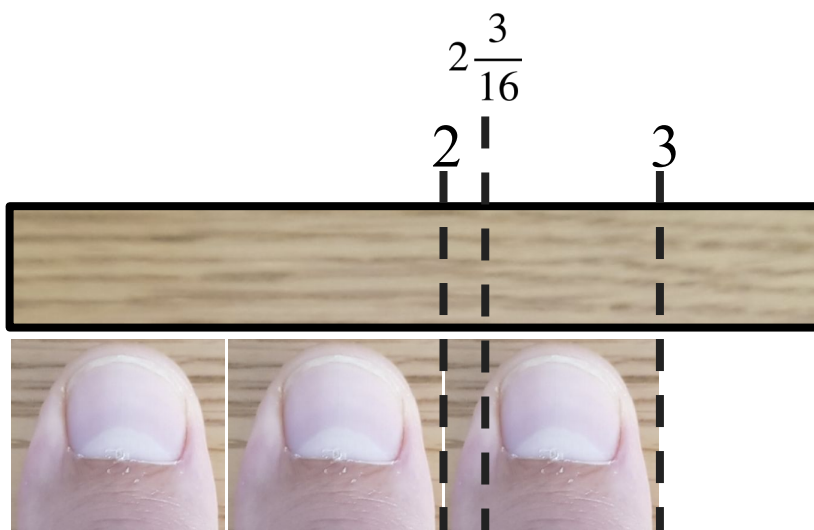


2) Use a rule of thumb to measure out the below:

a) One and three eighths of an inch.



b) Two and three sixteenths of an inch.



3) a) How many millimetres in a centimetre?

10

One centimetre

$$\frac{1}{100} \text{ m}$$

$$1 \times 10^{-2} \text{ m}$$

One millimetre

$$\frac{1}{1000} \text{ m}$$

$$1 \times 10^{-3} \text{ m}$$

b) How many micrometres in a millimetre?

One micrometre

$$\frac{1}{1\,000\,000} \text{ m}$$

$$1 \times 10^{-6} \text{ m}$$

$$(1 \times 10^{-3}) \div (1 \times 10^{-6}) = 1000$$

c) How many micrometres in a centimetre?

$$(1 \times 10^{-2}) \div (1 \times 10^{-6}) = 10\,000$$

d) Engineers can laser cut to a precision of  $2.5 \times 10^{-5} \text{ m}$   
What fraction of a centimetre is this?

$$\frac{2.5 \times 10^{-5}}{1 \times 10^{-2}} = \frac{2.5 \times 10^{-5} \times 10^5}{1 \times 10^{-2} \times 10^5} = \frac{2.5}{10^3} = \frac{25}{10\,000} = \frac{1}{400}$$

*Can you imagine one four-hundredth of a centimetre! This is the kind of precision engineers operate to.*

## Task B: Other applications of maths

1) Show whether each of these walls is upright.

